LMI Master Project Specification

Kamaleshwaram Hub Blueprint: Dialysis, Diagnostics, Male/Female Nursing Home & ABDM/PM-JAY Architecture

EXECUTIVE SUMMARY & UMBRELLA MISSION

LMI is established as a perpetual Section 8 charitable umbrella brand covering healthcare, education, and social welfare. Operating under the principle of *Sadaqah Jariyah*, the immediate priority is the development of the 50-cent Kamaleshwaram Hub in Thiruvananthapuram, featuring a G+6 multi-level clinical and residential facility.

1. SITE & BUILDING PROFILE (G+6 STRUCTURAL LAYOUT)

The 50-cent site along the Kamaleshwaram two-way bus route permits a G+6 (7 levels above ground + basement) structural footprint, providing ~50,000 sq. ft. of total built-up area while adhering to KMBR Group C / Group A2 regulations and avoiding high-rise complexity (<15m fire threshold per floor tier):

Floor Level	Primary Operational Function	Usable Floor Area	Key Infrastructure & Amenities
Basement	Utilities & Parking	~10,000 sq. ft.	Industrial RO Water Plant (500 LPH), ETP, Generator, UPS, Parking
Ground Floor	Triage & Diagnostics	~8,000 sq. ft.	Reception, ABHA Kiosks, Pharmacy, Pathology Lab, Ultrasound, ECG
1st Floor	Main Dialysis Hall	~8,000 sq. ft.	30–50 Hemodialysis Chairs, Central Nursing Desk, Reprocessing Suite
2nd Floor	Secondary Wards	~8,000 sq. ft.	Isolation Dialysis Bays (HBsAg/HCV positive), Special Care Unit
3rd Floor	Specialist OPD & Admin	~8,000 sq. ft.	Nephrology/General OPD Clinics, LMI Foundation Administrative Office
4th Floor	**Female Nursing Home**	~8,000 sq. ft.	25–30 Segregated Beds, Keycard Access, Nurse Station, CKD Pantry
5th Floor	**Male Nursing Home**	~8,000 sq. ft.	25–30 Segregated Beds, Attendant Suites, On-call Doctor Rooms
6th Floor	IT & CSR Headquarters	~8,000 sq. ft.	FastAPI Engine Server Room, ABDM Gateway, Conference & Training Hall

2. TURNKEY PROJECT FINANCIAL BREAKDOWN

Project Phase / Component	Scope & Specifications	Estimated Capital (INR)
1. Land Acquisition	50 Cents in Kamaleshwaram (includes 8% registration & legal clearance)	**₹10.80 Crores**
2. Civil Construction (G+6)	~50,000 sq. ft. total built-up area (@ ~₹2,200/sq. ft. KMBR compliant)	**₹11.00 Crores**
3. Clinical & RO Setup	10–20 Dialysis Machines, 500 LPH Medical RO Plant, Reprocessors, IT Infrastructure	**₹1.20 Crores**
4. Male/Female Nursing Home	Medical beds, partitions, air conditioning, furnishings, emergency alarms (4th & 5th Fl)	**₹0.80 Crores** *(₹80 Lakhs)*
5. Working Capital Buffer	60-day operational reserve for consumables, utility tariffs, and clinical salaries	**₹0.20 Crores** *(₹20 Lakhs)*
Total Turnkey Capital	**Full Ownership (Land + Building + Clinical Hub + Nursing Home)**	**~₹24.00 Crores**

3. FINANCIAL SUSTAINABILITY & CROSS-SUBSIDY FRAMEWORK

LMI operates a self-sustaining financial model that shields the facility from government reimbursement backlogs while ensuring zero destitute patient rejection:

THE 3-TIER PATIENT SUBSIDY MODEL

1. Government PM-JAY / KASP: Reimbursed at ₹1,500–₹1,800/session by SHA Kerala. Direct cost is ~₹400, leaving a ~₹1,100 surplus per session to fund operations.
2. General Self-Pay Category: Fixed non-profit rate of **₹500 per session**. Covers direct material consumables and brings daily cash liquidity to the trust.
3. Destitute Category: 100% Free treatment and nursing home stay, funded via PM-JAY surplus, CSR grants, and *Sadaqah Jariyah* reserves.

Revenue Stream	Target Category	Financial Role in LMI Engine
PM-JAY Reimbursements	BPL / KASP Card Holders / 70+ Seniors	Yields ₹1,100 net surplus per session; acts as deferred capital expansion bonus.
Subsidized Rate (₹500)	General / Uncovered Patients	Provides instant daily cash liquidity to purchase daily consumables & pay utilities.
Nursing Home Fees	Attendants & Recovery Patients	Nominal ₹300–₹500/day fees cover floor maintenance & dietary support.
SIP & Reserve Yields	LMI Section 8 Trust Funds	Acts as an independent liquidity anchor, paying staff salaries during SHA payout delays.

4. TECHNICAL INTEGRATION DOMAIN & ENDPOINT MATRIX

All digital integrations operate on self-hosted infrastructure on Oracle Cloud Always Free Tier using `khagatara.com` domain routing:

ORACLE CLOUD ALWAYS FREE DEPLOYMENT ARCHITECTURE

- **Compute:** Ampere A1 ARM Instance (2 OCPUs, 12 GB RAM, 200 GB Storage) — **₹0/month**.
- **Subdomain Mappings:** `khi.khagatara.com` (Web Portal) | `abdm.khagatara.com` (NHA Webhook Bridge) | `api.khagatara.com` (FastAPI Engine).
- **Security:** NGINX Reverse Proxy with automated Let's Encrypt TLS v1.3 SSL renewals.

5. ABDM & PM-JAY FASTAPI CORE GATEWAY CODE

Production Python service deployed on Oracle Cloud for session token caching and asynchronous webhook handling:

```
import os, requests, time
from fastapi import FastAPI, Request, HTTPException
from pydantic import BaseModel

app = FastAPI(title="LMI Health Engine", version="1.0.0")

ABDM_CLIENT_ID = os.getenv("ABDM_CLIENT_ID", "LMI_SANDBOX_ID")
ABDM_CLIENT_SECRET = os.getenv("ABDM_CLIENT_SECRET", "LMI_SANDBOX_SECRET")
AUTH_URL = "https://dev.abdm.gov.in/gateway/v0.5/sessions"

token_cache = {"token": None, "expires_at": 0}

def get_abdm_access_token():
    if token_cache["token"] and time.time() < token_cache["expires_at"]:
        return token_cache["token"]

    payload = {"clientId": ABDM_CLIENT_ID, "clientSecret": ABDM_CLIENT_SECRET}
    res = requests.post(AUTH_URL, json=payload, headers={"Content-Type": "application/json"})
    if res.status_code == 200:
        data = res.json()
        token_cache["token"] = data["accessToken"]
        token_cache["expires_at"] = time.time() + data.get("expiresIn", 1400)
        return token_cache["token"]
    raise HTTPException(status_code=500, detail="ABDM Authentication Failed")

@app.post("/v0.5/users/auth/on-init")
async def abha_auth_on_init(request: Request):
    body = await request.json()
    return {"status": "ACKNOWLEDGED", "requestId": body.get("requestId")}
```

6. NHCX PM-JAY CLAIM SUBMISSION ENGINE (PACKAGE KMC001)

Automated submission logic for generating cashless hemodialysis claim bundles for SHA Kerala:

```
@app.post("/api/v1/nhcx/submit-claim")
def submit_pmjay_claim(patient_abha: str, pmjay_card_id: str, package_code: str = "KMC001"):
    token = get_abdm_access_token()
    claim_payload = {
        "resourceType": "Claim",
        "identifier": [{"system": "https://nhcx.gov.in/claim", "value": f"LMI-CLAIM-{int(time.time())}"}],
        "status": "active",
        "type": {"coding": [{"system": "http://terminology.hl7.org/CodeSystem/claim-type", "code": "instit
utional"}]},
        "use": "claim",
        "patient": {"identifier": {"system": "https://healthid.abdm.gov.in", "value": patient_abha}},
        "provider": {"identifier": {"system": "https://hpr.abdm.gov.in", "value": "LMI_TVM_01"}},
        "insurance": [{"sequence": 1, "focal": True, "coverage": {"display": f"PMJAY Card: {pmjay_card_i
d}"}}],
        "item": [{
            "sequence": 1,
            "productOrService": {
                "coding": [{"system": "https://pmjay.gov.in/packages", "code": package_code, "display": "H
emodialysis Session"}]
            },
            "unitPrice": {"value": 1800, "currency": "INR"}
        }]
    }
    return {"status": "SUBMITTED", "claim_id": claim_payload["identifier"][0]["value"], "payload": claim_p
ayload}
```

7. IMPLEMENTATION ROADMAP & MILESTONE CHECKLIST

Phase	Key Milestone Target	Regulatory / Operational Body	Target Completion
Phase 1	Section 8 Company Incorporation ("LMI Foundation")	Ministry of Corporate Affairs (MCA)	Month 1
Phase 2	50-Cent Land Acquisition / 99-Year Lease Execution	Revenue Dept / Private Landowner	Month 2–3
Phase 3	KMBR Building Permit Approval (G+6 Layout)	Thiruvananthapuram Corporation	Month 4–5
Phase 4	ABDM Sandbox Exit & Production API Authorization	National Health Authority (NHA)	Month 6
Phase 5	Civil Construction Completion & Nursing Home Fit-out	LMI Projects Division	Month 14
Phase 6	PM-JAY / KASP Hospital Empanelment & Go-Live	State Health Agency (SHA Kerala)	Month 16

8. KEY COMPLIANCE & REGULATORY GUIDELINES

- **KMBR Group C & A2 Compliance:** Maintain mandatory 6m front setback and 3m–5m side/rear yards for fire tender access around the building perimeter.
- **Pollution Control Board (KPCB) Clearance:** Operationalize an on-site Effluent Treatment Plant (ETP) specifically calibrated for dialysis fluid disposal prior to clinical commissioning.
- **12A & 80G Income Tax Exemptions:** File immediately following Section 8 incorporation to allow corporate donors to route CSR funds to LMI tax-free.
- **DISHA / ABDM Privacy Norms:** Ensure all patient clinical records are encrypted locally with AES-256 GCM before syncing to cloud endpoints.

FINAL PROJECT AUTHORIZATION SIGN-OFF

This master specification serves as the formal operational, architectural, and technical blueprint for the **LMI Hub at Kamaleshwaram**. It is ready for presentation to legal counsel, municipal architects, and government empanelment boards.